

PYRAMID QUESTIONS

1) Draw the following pyramids of numbers.

- i) 15 grass plants → 12 Crickets → 1 Magpie
- ii) 5 Wattle Bushes → 200 termites → 3 lizards
- iii) 1 Gum tree → 2000 termites → 5 Kookaburra's

a) Which one looks odd? Why does it have this shape?

b) Use the following information to draw a pyramid of biomass for each food chain

Organism	Weight (each)	Organism	Weight (each)	Organism	Weight (each)
Grass	10g	Crickets	5g	Magpie	20g
Wattle bush	250g	Termites	0.5g	Lizard	10g
Gum tree	1000Kg	Termites	0.5g	Kookaburra	10g

c) Do any of them look odd? Why not?

2) The following information was collected by a class of pupils studying the numbers of animals feeding on an oak tree.

Animal	Number	Food which each animal eats
Moth larvae	93	Oak leaves
Chiffchaff	6	Moth larvae
Vole	3	Oak fruits
Sparrowhawk	2	Chiffchaffs and blue tits
Blue tit	3	Moth larvae
Carabid beetle	5	Moth larvae

Copy the following table and use the information above to complete it.

Trophic level	Names of organisms	Number of organisms	Biomass
Tertiary consumers			1
Secondary consumers			6.6
Primary consumers			12.5
Producers			177

- a) Draw a pyramid of numbers and a pyramid of biomass – to scale - for the above results.
- b) Why might the information collected by the children not give a completely accurate picture of the number of organisms in the oak tree?